Specifically, if the maximal value of the objective function in (4a) is not smaller than R^L, then the RBs within $\mathcal{N}$ can meet the constraints of problem (3), i.e., problem (3) is feasible. Otherwise, the RBs within $\mathcal{N}$ are insufficient, and problem (3) is infeasible.

1) Optimal Power Allocation: We first study the optimal power allocation of problem (4). By introducing Lagrangian multipliers η and ν for constraints (4b) and (4c), respectively, and considering the positive power constraint in (3h), we can obtain the optimal power allocation from the Karush-Kuhn-Tucker (KKT) conditions (the details are omitted for the lack of space) as

$$p_f = \frac{x_f^L + x_f^S \eta}{\nu} - \frac{1}{\gamma_f}. \tag{5}$$

The power allocation policy in (5) has a two water-level structure. For the RBs allocated to the LBT, i.e., $\{\text{RB}_f | x_f^L = 1, x_f^S = 0, f \in \mathcal{N}\}$, (5) reduces to $p_f = \frac{1}{\nu} - \frac{1}{\gamma_f}$. The water-level is $\frac{1}{\nu}$, which should satisfy

$$\frac{1}{\nu} > \frac{1}{\gamma_{\min}^L} \tag{6}$$

to ensure the positive power constraint in (3h), where $\gamma_{\min}^L$ denotes the minimum of γ_f for the RBs allocated to the LBT. Similarly, for the RBs allocated to the SBT, i.e., $\{\text{RB}_f | x_f^L = 0, x_f^S = 1, f \in \mathcal{N}\}$, (5) becomes $p_f = \frac{\eta}{\nu} - \frac{1}{\gamma_f}$. The water-level becomes $\frac{\eta}{\nu}$, which should satisfy

$$\frac{\eta}{\nu} > \frac{1}{\gamma_{\min}^S}, \tag{7}$$

where $\gamma_{\min}^S$ denotes the minimum of γ_f for the RBs allocated to the SBT.

The two water-levels can be derived by substituting (5) into (4b) and (4c), which are

$$\frac{1}{\nu} = \frac{P_{\max} + \sum\limits_{f\in\mathcal{N}} \frac{1}{\gamma_f} - N^S e^{\frac{1}{N^S}\left(\frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} - \sum\limits_{f\in\mathcal{N}} x_f^S \log\gamma_f\right)}}{N^L}, \tag{8a}$$

$$\frac{\eta}{\nu} = e^{\frac{1}{N^S}\left(\frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} - \sum\limits_{f\in\mathcal{N}} x_f^S \log\gamma_f\right)}, \tag{8b}$$

where $N^L = \sum_{f\in\mathcal{N}} x_f^L$ and $N^S = \sum_{f\in\mathcal{N}} x_f^S$ denote the total number of RBs allocated to the LBT and SBT, respectively, and $N^L + N^S = N$ with N denoting the total number of RBs in $\mathcal{N}$. It is clear that the water-levels depend on the RB allocation results x_f^L and x_f^S.

2) Optimal RB Allocation: By substituting the optimal power allocation in (5) and (8) into problem (4) and considering the constraints in (6) and (7) for water-levels, we can eliminate p_f and obtain an optimization problem only with respect to x_f^L and x_f^S. The objective function in (4a) can be derived as (9) (at the top of next page), where we define $X \triangleq \sum_{f\in\mathcal{N}} x_f^S \log\gamma_f$ and $Y \triangleq -X + N^L \log(P_{\max} + \sum_{f\in\mathcal{N}} \frac{1}{\gamma_f} - N^S e^{\frac{\frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} - X}{N^S}})$ for brevity.

The objective function in (9) is a complicated function of x_f^S (or equivalently x_f^L since $x_f^L + x_f^S = 1$ in (3e)), where the terms X, Y, N^S, and N^L are all functions of x_f^S. Thus, it is difficult to directly optimize x_f^S with (9). In the following we first optimize the RB allocation for any given N^S and then find the best value of N^S.

Given N^S, only X and Y in (9) are related to x_f^S. Then, by omitting the irrelevant terms, we can rewrite problem (4) as the following problem

$$\max_{x_f^S} Y \tag{10a}$$

$$\text{s.t.} X > \underbrace{\frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} - N^S \log\Big(\frac{1}{N^S}\big(P_{\max} + \sum\nolimits_{f\in\mathcal{N}} \frac{1}{\gamma_f} - \frac{N^L}{\gamma_{\min}^L}\big)\Big)}_{\triangleq X^{lb}}, \tag{10b}$$

$$X < \underbrace{\frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} + N^S \log\gamma_{\min}^S}_{\triangleq X^{ub}}, \tag{10c}$$

$$x_f^S \in \{0, 1\}, \tag{10d}$$

$$\text{(3d)},$$

where x_f^L is replaced by $1 - x_f^S$ according to (3e), (10b) and (10c) come from the water-level constraints in (6) and (7), and X^{lb} and X^{ub} denote the lower and upper bounds of X in (10b) and (10c), respectively.

In problem (10), the optimal RB allocation given N^S is to choose N^S RBs for the SBT, which can maximize Y meanwhile satisfying constraints in (10b)∼(10d) and (3d). To solve problem (10), let us first analyze the properties of the objective function Y.

Property 1: Y, defined in (9), is a concave function of X since $\nabla_X^2 Y < 0$. The maximum of Y is achieved at X^*,

$$X^* \triangleq \frac{R^S}{LB} + Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} + N^S \log\frac{N}{P_{\max} + \sum_{f\in\mathcal{N}} \frac{1}{\gamma_f}}, \tag{11}$$

which can be obtained by letting $\nabla_X Y = 0$. Herein, $\nabla_X^2 Y$ and $\nabla_X Y$ are the first and second derivatives of Y with respect to X.

Note that the value of X^* in (11) that maximizes Y may fall outside the feasible region of X in problem (10). This is because constraints in (10b) and (10c) limit X to the interval (X^{lb}, X^{ub}) and constraints in (10d) and (3d) restrict X (i.e., $\sum_{f\in\mathcal{N}} x_f^S \log\gamma_f$) to specific discrete values. Consequently, the optimal value of X has the following property.

Property 2:

- If $X^{ub} \le X^*$, Y increases with X within (X^{lb}, X^{ub}). The optimal X is the discrete value that is closest to X^{ub} within (X^{lb}, X^{ub}).
- If $X^{lb} < X^* < X^{ub}$, Y is concave for X. Let X_1 denote the discrete value that is closest to X^* within $(X^{lb}, X^*]$ and X_2 denote the discrete value that is closest to X^* within (X^*, X^{ub}). Then, the optimal X is X_1 if X_1 leads to a larger value of Y than X_2, or X_2 otherwise.

$$
\begin{aligned}
& LB\sum\nolimits_{f\in\mathcal{N}} x_f^L \log\left(\frac{\gamma_f}{N^L}\Big(P_{\max}+\sum\nolimits_{f\in\mathcal{N}}\frac{1}{\gamma_f}-N^S e^{\frac{\frac{R^S}{LB}+Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}}-\sum_{f\in\mathcal{N}} x_f^S\log\gamma_f}{N^S}}\Big)\right)\\
&= LB\log\left(\frac{\prod_{f\in\mathcal{N}}\gamma_f}{\prod_{f\in\mathcal{N}} x_f^S\gamma_f}\left(\frac{1}{N^L}\left(P_{\max}+\sum\nolimits_{f\in\mathcal{N}}\frac{1}{\gamma_f}-N^S e^{\frac{\frac{R^S}{LB}+Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}}-X}{N^S}}\right)\right)^{N^L}\right) \qquad (9)\\
&= LB\left(\log\prod\nolimits_{f\in\mathcal{N}}\gamma_f+Y-N^L\log N^L\right),
\end{aligned}
$$

- If $X^* \le X^{lb}$, Y decreases with X within (X^{lb}, X^{ub}). The optimal X is the discrete value that is closest to X^{lb} within (X^{lb}, X^{ub}).

Property 2 indicates that the optimal X has three possible values, which are the discrete value closest to X^{lb}, X^*, or X^{ub}. The following Property 3 indicates which case should be selected. The detailed derivations can be found in Appendix A.

Property 3:

$$
\begin{cases}
X^{lb} < X^* < X^{ub}, & \text{if } \gamma_{\min} > \psi,\\
X^* \le X^{lb}, & \text{if } \gamma_{\min} \le \psi, \gamma_{\min}^L = \gamma_{\min},\\
X^{ub} \le X^*, & \text{if } \gamma_{\min} \le \psi, \gamma_{\min}^S = \gamma_{\min},
\end{cases}
$$

where $\psi \triangleq \frac{N}{P_{\max}+\sum_{f\in\mathcal{N}}\frac{1}{\gamma_f}}$, $\gamma_{\min}$ is the minimal channel gain of the RBs in $\mathcal{N}$, and the condition $\gamma_{\min}^L = \gamma_{\min}$ or $\gamma_{\min}^S = \gamma_{\min}$ means that the RB with the minimal channel gain is allocated to the LBT or SBT, respectively.

By combining Property 2 and Property 3, the optimal RB allocation can be determined by examining the three conditions listed in Property 3. For example, if $\gamma_{\min} \le \psi$ and $\gamma_{\min}^L = \gamma_{\min}$, i.e., the second condition of Property 3 holds, then we have $X^* \le X^{lb}$. This corresponds to Case 3 of Property 2, which indicates that the optimal RB allocation x_f^S will make the optimal value of X closest to X^{lb} within (X^{lb}, X^{ub}). This leads to the following optimization problem

$$
\begin{aligned}
\min_{x_f^S} \quad & \sum\nolimits_{f\in\mathcal{N}} x_f^S \log\gamma_f - X^{lb} && \text{(12a)}\\
\text{s.t.} \quad & X^{lb} < \sum\nolimits_{f\in\mathcal{N}} x_f^S \log\gamma_f < X^{ub}, && \text{(12b)}\\
& \text{(10d), (3d).}
\end{aligned}
$$

Problem (12) aims to select N^S RBs from $\mathcal{N}$ such that $\sum_{f\in\mathcal{N}} x_f^S \log\gamma_f$ is closest to X^{lb}. This is a combinatorial optimization problem, which falls into the framework of "subset-sum problems" in the real-number domain. There exist efficient algorithms for solving the subset-sum problems, e.g., [25], which can be applied to solve problem (12).

We are now ready to present the whole algorithm to solve the original problem (2) where $M = 1$. The algorithm has a hierarchical structure. In the outer loop, we successively increase the number of RBs N with good channel gains until the QoS requirements of both LBT and SBT are met. In the inner loop, we search the number of RBs N^S allocated to the SBT for a given N, and solve a series of subset-sum problems based on Property 2 and Property 3. We then obtain the optimal RB allocation policy for any specified N^S, with which the optimal power allocation can be obtained from (5).

The detailed algorithm is summarized in Algorithm 1. In the outer loop, N is initialized by $N_{\min}$, which can be computed as follows. For any given number of RBs N, the maximal achievable rate of the user can be obtained by using the classic water-filling power allocation. If this maximal rate is lower than $R^L + R^S$, the available number of RBs N is insufficient to support both the LBT and SBT. Accordingly, we set the minimal number of RBs as $N_{\min} + 1$, ensuring the data rate not less than $R^L + R^S$.

Algorithm 1 Optimal power and RB allocation for single-user scenario

Input: Sort $\gamma_f, f = 1, \ldots, F$, in a descending order.
Output: x_f^L, x_f^S and $p_f, f \in \mathcal{N}$.

Initialization: $N = N_{\min}$ and $\mathcal{N}$ consists of N best RBs.
repeat
  Add an unselected RB with the largest γ_f to $\mathcal{N}$, and update $N \leftarrow N + 1$.
  for $N^S = 1, \ldots, N$ **do**
    if $\gamma_{\min} > \frac{N}{P_{\max}+\sum_{f\in\mathcal{N}}\frac{1}{\gamma_f}}$ **then**
      Choose case 2 of Property 2.
      Obtain x_f^S by solving the subset-sum problem with the method in [25].
      Compute the value of (9), denoted by $J(N^S)$.
    else
      let $\gamma_{\min}^L = \gamma_{\min}$, i.e., $x_N^S = 0$ **do**
        Choose case 3 of Property 2 and obtain x_f^S by solving the subset-sum problem.
        Compute the value of (9), denoted by J_1.
      let $\gamma_{\min}^S = \gamma_{\min}$, i.e., $x_N^S = 1$ **do**
        Choose case 1 of Property 2 and obtain x_f^S by solving the subset-sum problem.
        Compute the value of (9), denoted by J_2.
      do Compute $J(N^S) = \max\{J_1, J_2\}$.
    end if
  end for
until $\max\{J(N^S)\} \ge R^L$
Obtain $x_f^S, x_f^L = 1 - x_f^S, f \in \mathcal{N}$ corresponding to the maximum of $J(N^S)$.
Compute $p_f, f \in \mathcal{N}$ with (5) and (8).

B. Extended Algorithm for Multiuser Scenario

Extending the algorithm developed for the single-user scenario to the multiuser scenario is non-trivial. This is because

in the multiuser case, it is no longer feasible to determine the occupied set of RBs for each user by simply sorting them based on the channel gains in the outer loop. Here, we propose a heuristic method in which users sequentially select RBs with the best channel conditions from the unselected RBs. This method is described as follows.

i. Every user, whose QoS requirements have not been fully satisfied, sequentially chooses an unselected RB with the largest channel gain, and adds this RB to the set of selected RBs.
ii. With these selected RBs, each user employs the algorithm designed for the single-user scenario to optimize resource allocation and verifies whether the QoS requirements for both LBT and SBT are satisfied.
iii. Users who have met their QoS requirements will exit the iteration. The remaining users repeat step i and ii.

This method takes into account the QoS requirements of users by letting users exit the iteration once their QoS requirements are satisfied, meanwhile it ensures a fair allocation of RBs among users by the sequential selection process. Within each iteration, given the total number of RBs selected by each user, the RB and power allocation for each user can be solved by the proposed single-user algorithm.

Although the proposed numerical algorithms significantly reduce the computational complexity compared to exhaustive searching, these iterative algorithms are still unsuitable for real-time implementation. In the next section, we propose a deep learning based method to reduce inference complexity.

IV. Learning based Method for Resource Allocation

The primary difficulties of designing learning based method to solve problem (2) lie in two aspects. First, the policy to be learned involves discrete RB allocation, resulting in either vanishing or exploding gradient issues during the training of DNNs. Second, the SBT has stringent reliability requirements, requiring the learned policy to satisfy the QoS constraints with very high probability.

To tackle the difficulties, in this section we develop an adaptive smoothing method to facilitate the learning of discrete variables, and design a nonlinear penalty function that amplifies the penalties for slight constraint violations.

A. Problem Transformation

We begin with an equivalent transformation of problem (2) with the following considerations. In problem (2), we use $p_{m,f}$ to denote the power allocated by user m to RB_f, and use the discrete variables $x^L_{m,f}$ and $x^S_{m,f}$ to denote whether RB_f is allocated to the LBT or SBT for user m. However, the power allocation and RB allocation are related. For instance, whether a RB is allocated to the LBT of a user is determined by whether the power allocated to the LBT of the user on the RB is positive or not. Thus, we can make an equivalent transformation of problem (2) that only involves power allocation variables. Then, RB allocations can be computed with corresponding powers. Such a transformation can effectively reduce the solution space for learning by eliminating non-meaningful solutions, such as those cases where RBs are occupied yet no power is allocated or where power is allocated to non-occupied RBs, e.g., $x^L_{m,f} = 1$ but $p_{m,f} = 0$. The transformation is achieved by defining new variables for power allocation, $p^L_{m,f}$ and $p^S_{m,f}$, which denote the powers of user m allocated to the LBT and SBT on RB_f, respectively.

To ensure that each RB is allocated to either the LBT or SBT of a single user, as constrained by (2d), at most one of power allocation variables $p^s_{m,f}$, for $s \in \{L, S\}$ and $m = 1, \ldots, M$, can be positive for RB_f, where $s \in \{L, S\}$ denotes LBT or SBT. To express this restriction, we introduce a piecewise maximal function $g(\cdot)$ as

$$g^s_{m,f} \triangleq g\big(p^s_{m,f}, \mathcal{P}^s_{m,f}\big) = \begin{cases} p^s_{m,f}, & p^s_{m,f} > \max \mathcal{P}^s_{m,f}, \\ 0, & \text{else}, \end{cases} \tag{13}$$

where $\mathcal{P}^s_{m,f} = \{p^i_{j,f} \mid i \neq s \text{ or } j \neq m, i \in \{L, S\}, j = 1, ..., M\}$ denotes all the power allocation variables to RB_f except for variable $p^s_{m,f}$. The function $g(\cdot)$ ensures that for $g^s_{m,f}, s \in \{L, S\}, m = 1, \ldots, M$, at most one $g^s_{m,f}$ is positive while the rest are zero, thereby satisfying constraints in (2d).

With (13), the final power allocation of user m on RB_f can be obtained as $g^s_{m,f}$, and the final RB allocation decisions can be obtained as $\mathbf{1}(g^s_{m,f})$, where $s \in \{L, S\}$ and $\mathbf{1}(\cdot)$ is an indicator function that outputs 1 for positive inputs and 0 for zero input. Then, we can reformulate problem (2) as

$$\min_{p^L_{m,f}, p^S_{m,f}} \sum_{f=1}^{F} \mathbf{1}(g_f) \tag{14a}$$

$$\text{s.t.} LB \sum_{f=1}^{F} \log\left(1 + \gamma_{m,f} g^L_{m,f}\right) \geq R^L_m, m = 1, \ldots, M, \tag{14b}$$

$$LB \sum_{f=1}^{F} \log\left(1 + \gamma_{m,f} g^S_{m,f}\right) - \sqrt{\frac{N^S_m LB}{\tau}} Q^{-1}(\epsilon_m) \geq R^S_m, \quad m = 1, \ldots, M, \tag{14c}$$

$$\sum_{f=1}^{F} \left(p^L_{m,f} + p^S_{m,f}\right) \leq P_{\max}, m = 1, \ldots, M, \tag{14d}$$

$$p^L_{m,f}, p^S_{m,f} \geq 0, f = 1, \ldots, F, m = 1, \ldots, M, \tag{14e}$$

where $g_f \triangleq \sum_{m=1}^{M} g^L_{m,f} + g^S_{m,f}$ denotes the total power allocated to RB_f by all users and $N^S_m = \sum_{f=1}^{F} \mathbf{1}(g^S_{m,f})$ denotes the total number of RBs allocated to the SBT of user m.

Problem (14) only optimizes the power allocations $p^L_{m,f}$ and $p^S_{m,f}$, where the two discrete functions $g(\cdot)$ and $\mathbf{1}(\cdot)$ ensure that the constraints in (2d) and (2e) are satisfied. As analyzed before, the introduced discrete functions $g(\cdot)$ and $\mathbf{1}(\cdot)$ can equivalently express the discrete RB allocation variables in problem (2), making problem (2) and (14) have the same optimal solution. Although such a transformation still faces the gradient vanishing or explosion issues during training, it allows us to design smoothing methods to approximate the discrete functions.

Following the learning framework for constrained optimization proposed by [20], problem (14) can be transformed into a primal-dual form by introducing Lagrangian multipliers